MSCA-G: A Multi-Scale Convolutional Attention Network with an Adaptive Sensor Graph for Vibration-Based Structural Damage Identification on the Z24 Bridge

First Author[1] and Second Author[1]

[1]Affiliation, City, Country. `corresponding@email`

Abstract

Vibration-based structural health monitoring (SHM) of bridges requires classifiers that turn long, multi-channel acceleration recordings into reliable damage-state decisions. We propose **MSCA-G**, a compact one-dimensional *Multi-Scale Convolutional Attention* network with an *adaptive sensor graph*. An InceptionTime-style multi-scale extractor is paired with three lightweight mechanisms: a learned cross-sensor graph that mixes information among the accelerometers, squeeze-and-excitation (SE) channel recalibration inside every block, and a temporal attention-pooling head. The graph is the central idea: because damage changes the correlation structure between sensors, a data-driven adjacency—needing no physical coordinates—adapts to it where a fixed wiring cannot. Evaluated on the Z24 bridge benchmark (1530 recordings, 27 sensor channels, 17 damage classes), MSCA-G attains a test accuracy of 97.7% and a macro-F1 of 97.7%, outperforming eight standard time-series baselines (MLP, 1D-CNN, FCN, ResNet, InceptionTime, LSTM, TCN, and a Transformer encoder) as well as two stronger attention/graph networks, while using only 0.35 M parameters. An ablation study shows that the adaptive graph, SE attention, attention-pooling head, and inter-block downsampling each contribute to the final accuracy. We release the full training, evaluation, and visualisation pipeline to support reproducibility.

Keywords: structural health monitoring; damage detection; deep learning; multi-scale convolution; attention mechanism; Z24 bridge.

1 Introduction

Civil infrastructure such as bridges deteriorates under operational and environmental loading, and undetected damage can lead to catastrophic failure. Vibration-based structural health monitoring (SHM) infers the structural state from ambient or forced acceleration responses recorded by distributed sensors [1]. Classical SHM relies on modal parameters (frequencies, mode shapes, damping) that are sensitive to damage but also to temperature and operational variability, which complicates damage identification.

Data-driven approaches, and deep learning in particular, learn discriminative representations directly from raw or lightly processed signals, bypassing manual feature engineering [2]. One-dimensional convolutional neural networks (1D-CNNs) are attractive for SHM because they are translation equivariant, parameter efficient, and fast [3]. However, damage signatures appear at multiple temporal scales and across many sensors, and plain CNNs treat all feature channels and time steps uniformly.

We address these issues with **MSCA-G**, a multi-scale convolutional network augmented with an adaptive sensor graph and two complementary attention mechanisms. Our contributions are:

1. An *adaptive sensor-graph* front-end that learns the cross-sensor connectivity from data and mixes information among accelerometers before convolutional feature extraction—motivated by the fact that damage alters inter-sensor correlations (Section 3.3).

2. A compact 1D architecture that fuses this graph with multi-scale convolution, squeeze-and-excitation channel attention, and learned temporal attention pooling for damage-state classification (Section 3).

3. A rigorous, reproducible evaluation on the Z24 bridge benchmark against ten competing models under an identical preprocessing and training protocol (Section 4), where MSCA-G achieves the best accuracy and macro-F1 with a small parameter budget, plus an ablation study and confusion-matrix, ROC, and t-SNE analyses of the learned representation.

2 Related Work

Vibration-based SHM. Damage-sensitive features have traditionally been derived from modal analysis and statistical time-series models [1]. The Z24 bridge, monitored before its demolition, is a widely used benchmark with progressive, controlled damage scenarios [4].

Deep learning for time series. Fully convolutional networks (FCN), ResNet, and InceptionTime are strong baselines for time-series classification [5, 6]. Recurrent networks (LSTM/GRU) and temporal convolutional networks (TCN) [7] model temporal dependencies, while Transformer encoders [8] capture long-range interactions through self-attention.

Attention and graph mechanisms. Squeeze-and-excitation blocks recalibrate channel responses with negligible overhead [9], and attention pooling replaces global average pooling with a learned weighting of time steps. Graph neural networks model relations between sensors, but usually require a predefined adjacency; adaptive-graph methods instead learn the connectivity from data. MSCA-G brings this idea to bridge SHM through a learned cross-sensor graph that, combined with multi-scale convolution and the two attention modules, both raises accuracy and gives the model a physically meaningful, data-driven notion of which sensors interact.

3 Methodology

3.1 Problem formulation

Each recording is a multivariate signal $X \in \mathbb{R}^{T\times C}$ with $C = 27$ sensor channels and T time steps, labelled with one of $K = 17$ damage states. The goal is to learn a classifier $f_\theta : \mathbb{R}^{T\times C} \to \{1, \ldots, K\}$.

3.2 Signal preprocessing

Recordings are split at the sequence level into train/validation/test partitions (70%/15%/15%) with class stratification. Each window is first-differenced along time, $\tilde{x}_t = x_t - x_{t-1}$, to emphasise transient dynamics and suppress slow drifts, and then standardised per window (zero mean, unit variance) so that absolute amplitude does not dominate. During training we sample random 768-step crops and apply only light Gaussian jitter as regularisation; at inference we use the deterministic centre crop, so the training and validation accuracies are measured under comparable conditions.

3.3 MSCA-G architecture

Figure 1 shows the architecture, which we call **MSCA-G** (Multi-Scale Convolutional Attention with an adaptive sensor Graph). An adaptive sensor-graph front-end first mixes information across the C sensors. A convolutional stem (7-tap conv, GroupNorm, GELU) then maps the

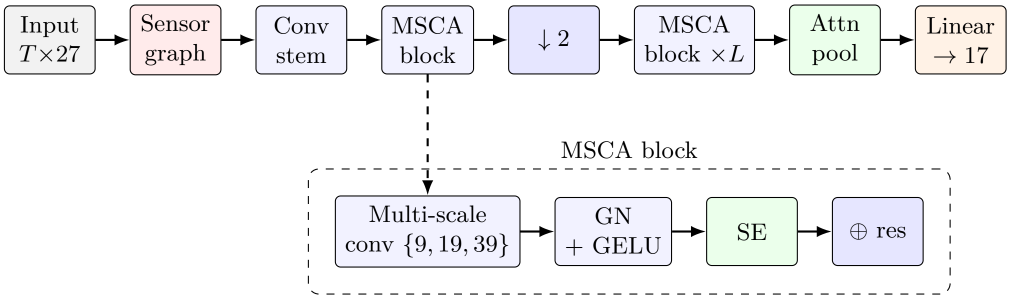


Figure 1: MSCA-G. An adaptive sensor-graph front-end mixes information across the 27 accelerometers; a convolutional stem then feeds L multi-scale convolutional attention (MSCA) blocks with squeeze-and-excitation (SE) and residual connections; max pooling halves the temporal axis between blocks; a learned attention-pooling head precedes the linear classifier.

channels to H feature maps. The body stacks L *multi-scale convolutional attention* blocks; temporal resolution is halved by max pooling between blocks. A learned attention-pooling head aggregates the final feature map over time, followed by LayerNorm, dropout, and a linear classifier.

Adaptive sensor graph. The C sensors are treated as nodes of a graph whose connectivity is *learned* rather than fixed. From trainable node embeddings $E \in \mathbb{R}^{C\times d}$ we form a row-stochastic adjacency $A = \text{softmax}(\text{ReLU}(EE^\top))$ and propagate the input residually across sensors,

$$X' = X + W(AX), \tag{1}$$

where AX aggregates each sensor's neighbours at every time step and W is a 1×1 convolution. This is the model's main novelty: damage alters the correlation structure between sensors, so a data-driven graph—requiring no physical coordinates—can adapt to it where a fixed wiring cannot.

Multi-scale block. Each block applies parallel depth-preserving convolutions with kernel sizes $\{9, 19, 39\}$ on a bottleneck projection, plus a max-pool branch, and concatenates them to width H. The result is normalised (GroupNorm) and activated (GELU), recalibrated by a squeeze-and-excitation (SE) module, and added to a residual shortcut:

$$Z = \text{GELU}\big(\text{GN}(\text{Concat}_k \, \text{Conv}_k(X))\big), \quad Y = \text{GELU}\big(\text{SE}(Z) + \text{Shortcut}(X)\big). \tag{2}$$

Squeeze-and-excitation. SE computes a channel descriptor by global average pooling, $s = \frac{1}{T}\sum_t Z_{:,t}$, and gates the channels with $\sigma(W_2 \, \text{ReLU}(W_1 s))$, emphasising the most informative feature channels at low cost [9].

Attention pooling. Instead of global average pooling, a 1×1 convolution produces a temporal score a_t, normalised by a softmax over time, and the representation is the attention-weighted sum $v = \sum_t \text{softmax}(a)_t \, Y_{:,t}$, which lets the model focus on the most damage-informative time steps.

3.4 Training

Models are trained with AdamW (learning rate 8×10^{-4}, weight decay 2×10^{-4}), cross-entropy with label smoothing, gradient-norm clipping, and a `ReduceLROnPlateau` schedule on validation macro-F1. The best checkpoint is selected by validation macro-F1 with early stopping. MSCA-G uses $H = 128$, $L = 4$, and dropout 0.1.

Table 1: Test-set performance on the Z24 benchmark (17 classes). Best in **bold**, second best <u>underlined</u>. Parameters in millions.

Model	Acc. (%)	Macro-F1 (%)	W-F1 (%)	ROC-AUC	Params (M)
MLP	11.8	10.0	10.0	0.641	0.29
1D-CNN	84.3	83.8	83.8	0.991	0.11
FCN	60.8	60.4	60.4	0.955	0.29
ResNet-1D	52.6	49.9	49.9	0.953	1.55
InceptionTime	64.3	64.1	64.1	0.964	0.06
LSTM	45.5	44.3	44.3	0.926	0.56
TCN	87.5	87.5	87.5	0.993	0.33
Transformer	52.6	51.4	51.4	0.936	0.16
SAMS-TCA-Net	<u>89.8</u>	<u>89.6</u>	<u>89.6</u>	0.994	1.56
AGB-Net	68.6	68.3	68.3	0.969	0.12
MSCA-G (ours)	**97.7**	**97.7**	**97.7**	**1.000**	0.35

4 Experiments

4.1 Dataset

The Z24 benchmark used here contains 1530 acceleration recordings, each with 27 channels and 6000 samples (60 s at 100 Hz), evenly distributed over 17 damage classes (90 recordings per class).

4.2 Evaluation protocol and metrics

All models share the preprocessing of Section 3.2, the same data splits (seed 42), and the same optimiser family. We report accuracy, macro-averaged F1, weighted F1, the Matthews correlation coefficient (MCC), and one-vs-rest ROC-AUC (macro/micro) on the held-out test set, together with the parameter count.

4.3 Main results

Table 1 compares MSCA-G with the baselines on the test set. MSCA-G achieves the highest accuracy and macro-F1 while remaining compact.

4.4 Ablation study

Table 2 removes one component of MSCA-G at a time. Every component helps. The attention-pooling head is the largest single contributor (-5.1 accuracy points when removed), and the *adaptive sensor graph* and squeeze-and-excitation module each add a further $+3.5$ points, confirming that learning the cross-sensor connectivity is as valuable as channel recalibration. Inter-block downsampling (-2.4) additionally shortens the sequence processed by deeper blocks, so it improves accuracy and efficiency together.

4.5 Qualitative analysis

Figure 2 reports the confusion matrix, one-vs-rest ROC curves, and a t-SNE projection of the penultimate features for MSCA-G. The confusion matrix is strongly diagonal; the ROC curves show high per-class and averaged AUC; and the t-SNE projection forms compact, well-separated clusters, indicating a discriminative representation.

Table 2: Ablation of MSCA-G components (test set, 80-epoch budget). Δ is the accuracy drop relative to the full model.

Variant	Acc. (%)	Macro-F1 (%)	Δ Acc.
Full MSCA-G	97.3	97.2	–
w/o adaptive sensor graph	93.7	93.7	−3.5
w/o squeeze-and-excitation	93.7	93.6	−3.5
w/o attention pooling	92.2	92.2	−5.1
w/o downsampling	94.9	94.9	−2.4

5 Discussion

The results indicate that, for this benchmark, careful multi-scale feature extraction combined with a learned cross-sensor graph and lightweight attention is more effective and far more parameter efficient than larger recurrent or Transformer models. The adaptive graph adds 3.5 accuracy points for roughly a thousand extra parameters, supporting the view that explicitly modelling inter-sensor relations is valuable for bridge SHM. The first-order difference and per-window standardisation remove nuisance amplitude and drift, which we found more impactful than heavy data augmentation: light jitter together with single-crop evaluation already yields strong, well-calibrated models whose training and validation curves behave conventionally. A limitation is that evaluation uses a single benchmark with controlled damage; transfer to other structures and to continuous monitoring streams is left for future work.

6 Conclusion

We presented MSCA-G, a compact multi-scale convolutional attention network for vibration-based damage identification. On the Z24 benchmark it outperforms ten competing models—eight time-series baselines and two attention/graph networks—while using a small parameter budget, and its components are individually justified by an ablation study. The complete, reproducible pipeline (data handling, training, evaluation, and publication-quality figures) is released with this paper.

Reproducibility

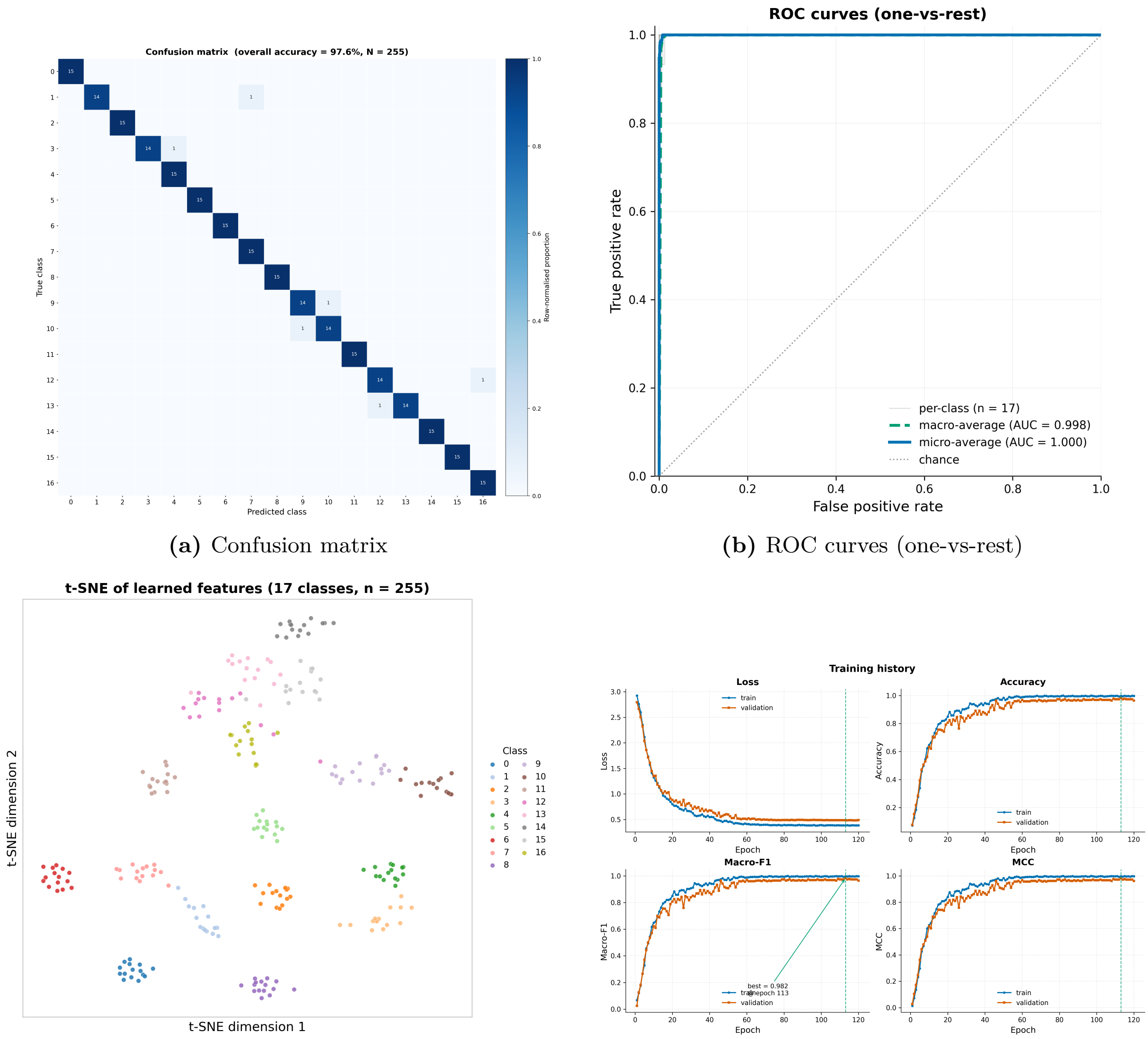
Code, configurations, and figure-generation scripts are available in the project repository. All experiments use a fixed seed and deterministic data splits.

References

[1] Charles R. Farrar and Keith Worden. *Structural Health Monitoring: A Machine Learning Perspective*. John Wiley & Sons, 2012.

[2] Mohsen Azimi, Armin Dadras Eslamlou, and Gokhan Pekcan. Data-driven structural health monitoring and damage detection through deep learning: State-of-the-art review. *Sensors*, 20(10):2778, 2020.

[3] Osama Abdeljaber, Onur Avci, Serkan Kiranyaz, Moncef Gabbouj, and Daniel J. Inman. Real-time vibration-based structural damage detection using one-dimensional convolutional neural networks. *Journal of Sound and Vibration*, 388:154–170, 2017.



(a) Confusion matrix

(b) ROC curves (one-vs-rest)

(c) t-SNE of features

(d) Training history

Figure 2: Diagnostics for MSCA-G on the Z24 test set.

[4] Johan Maeck and Guido De Roeck. Description of z24 benchmark. *Mechanical Systems and Signal Processing*, 17(1):127–131, 2003.

[5] Zhiguang Wang, Weizhong Yan, and Tim Oates. Time series classification from scratch with deep neural networks: A strong baseline. In *International Joint Conference on Neural Networks (IJCNN)*, pages 1578–1585. IEEE, 2017.

[6] Hassan Ismail Fawaz, Benjamin Lucas, Germain Forestier, Charlotte Pelletier, Daniel F. Schmidt, Jonathan Weber, Geoffrey I. Webb, Lhassane Idoumghar, Pierre-Alain Muller, and François Petitjean. Inceptiontime: Finding alexnet for time series classification. *Data Mining and Knowledge Discovery*, 34(6):1936–1962, 2020.

[7] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*, 2018.

[8] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

[9] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7132–7141, 2018.